

INNOVATIVE MINING SYSTEMS

Deep sea mining

By

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INTRODUCTION

- Deep-sea mining is an emerging industrial process designed to retrieve vast mineral deposits from the ocean floor, typically at depths greater than 200 m. These deposits contain "transition-critical" metals—like cobalt, nickel, and manganese—that are essential for the production of electric vehicle (EV) batteries, wind turbines, and other green energy technologies.

- Polymetallic Nodules (Manganese Nodules)

These are potato-sized, rock-like clumps found scattered across the deep-water abyssal plains

Key Minerals: Manganese (28%), Nickel (1.3%), Copper (1.1%), and Cobalt (0.2%).

Other Elements: Trace amounts of Rare Earth Elements (REEs), molybdenum, yttrium, and platinum.

Formation: They precipitate slowly from seawater over millions of years around a nucleus like a shark's tooth or rock fragment.

- Seafloor Massive Sulfides (SMS)

These deposits form around hydrothermal vents—"black smokers"—along mid-ocean ridges where superheated, mineral-rich water hits the cold ocean, causing minerals to precipitate.

Key Minerals: Rich in Copper, Gold, Silver, and Zinc.

Other Elements: Lead, barium, and palladium.

Location: Found at shallower depths (1,000–5,000 m) compared to nodules

INTRODUCTION

➤ Cobalt-Rich Ferromanganese Crusts

These crusts form "carpets" on the surfaces of underwater mountains known as seamounts, particularly in the Western Pacific.

Key Minerals: High concentrations of Cobalt (up to 2%), plus Manganese and Iron.

Other Elements: Platinum, tellurium, nickel, and REEs.

Challenge: They are firmly attached to bedrock, making them more difficult to extract than loose nodules.

Due to depletion of mineral resources, researchers and scientists have found an alternative, they have prospected mineral resources in ocean.

Many countries are exploring and exploiting minerals from the ocean.

We are going to discuss what are the minerals found in ocean.

Types of marine mining:

Seabed mining can be subdivided into two components:

- Shallow mining
- Deep sea mining

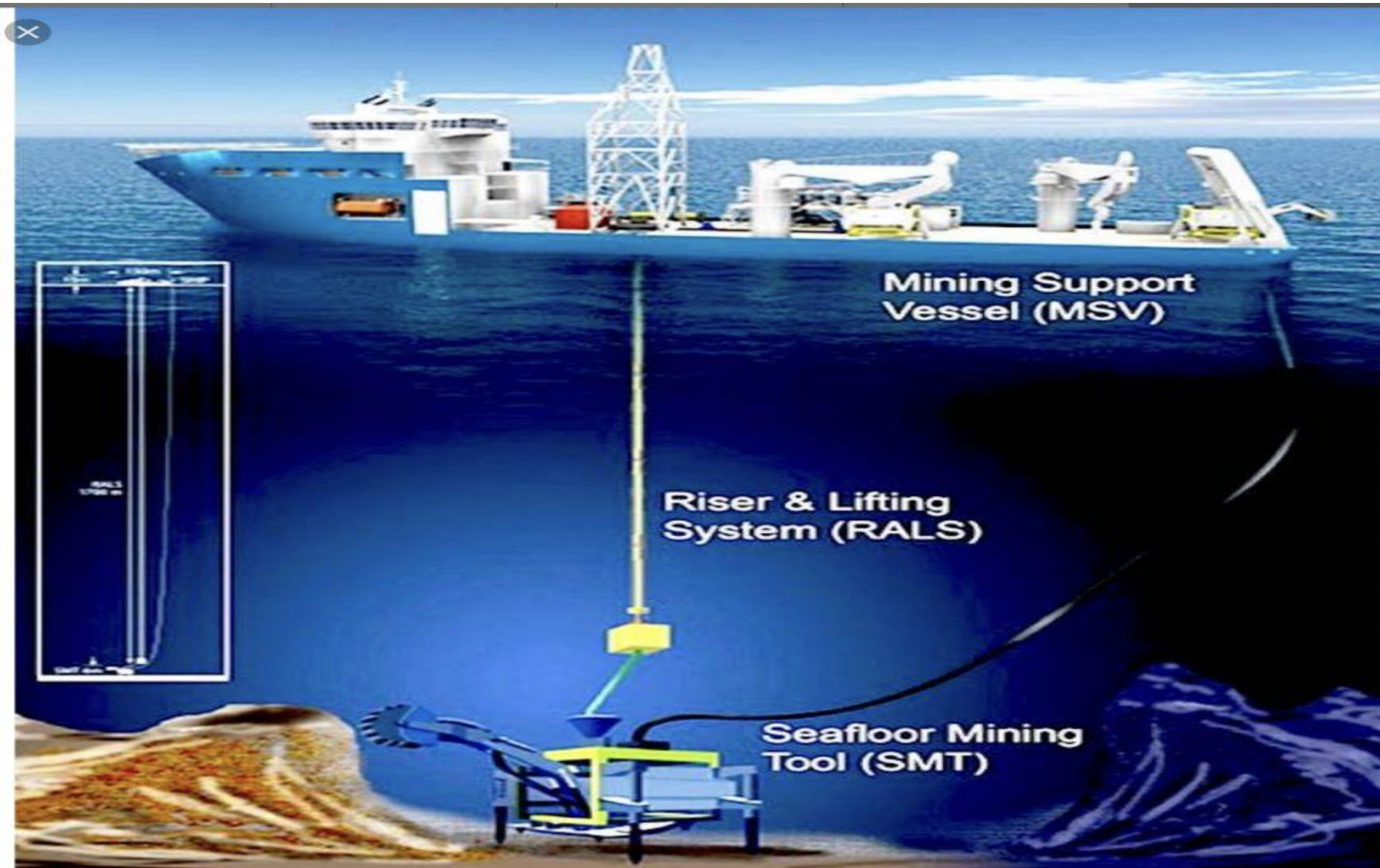
SHALLOW MARINE MINING

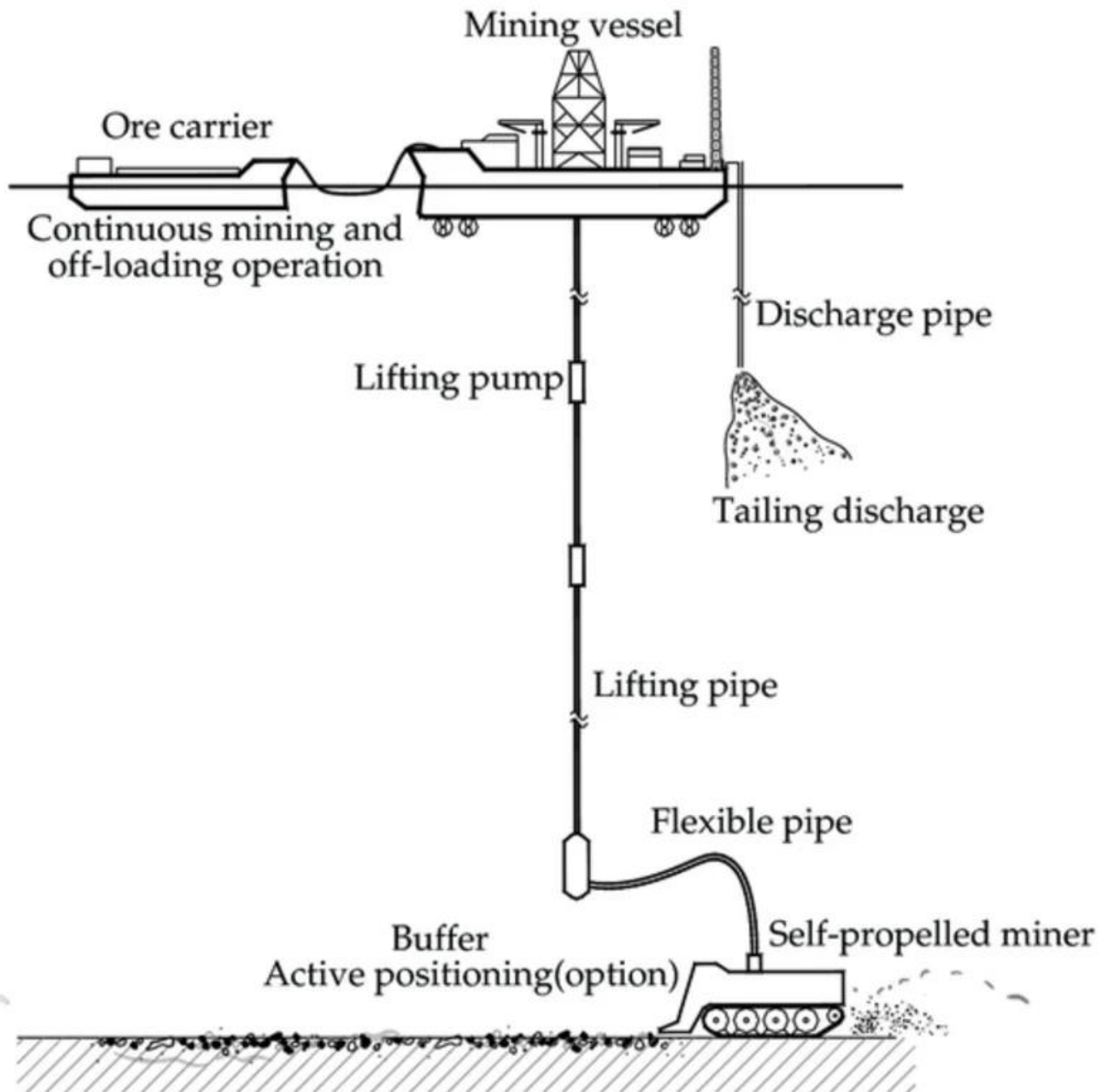
- Shallow marine mining largely refers to the extraction of mud, sand and gravel for construction purposes.
- Also refer to the mining of valuable minerals in the nearshore shallow waters.
- Construction sand and precious stones (gemstones) in shallow marine mining areas has been ongoing in various parts.
- Such mining is common in the pacific islands region and the costal area of southern Africa.

DEEP SEA MINING

- Deep sea mining is relatively new, extreme and dangerous mineral retrieval process that takes on the ocean floor.
- Ocean mining sites are usually around large areas of polymetallic nodules at about 1400 to 3700 m below the ocean surface.
- The vents create sulphide deposits, which contain certain valuable metals such as silver, gold, copper, manganese, cobalt, and zinc.

DEEP SEA MINING





MINING TECHNOLOGY DEVELOPMENT

- Since the early 1990s, countries including Germany, Japan, China, Korea, India and United States, were reported have been working on developing new offshore mining systems for collecting and lifting nodules.
- Shallow marine mining industry, gas and oil production technologies are transferable to deep sea bed mining.

➤ The planned mining system has three major components:

a) Mining Support Vessel (MSV)

b) Riser and Lifting System (RALS)

c) Seafloor Mining Tool (SMT)

The system is designed to operate at production rate of 6000 tons of Seafloor Massive Sulphide (SMS) ore a day at Nautilus minerals.

EFFECTS OF MARINE MINING ON ENVIRONMENT

- Marine mining is a relatively new field, the complete consequences of full scale mining operation are unknown. But it is believed that:
 - Removal of parts of sea floor will result in disturbances to the benthic layer (the bottom-most zone of a water body).
 - Increased toxicity of the water column and sediment plumes from tailings.
 - Removing parts of sea floor to disturb the habitat of benthic organisms, with unknown long time effects.

EXTRACTION

- Recent technological advancements have given rise to the use remotely operated vehicles (ROVs) to collect mineral samples from prospective mine sites.
- Using drills and other cutting tools, the ROVs obtain samples to be analyzed for precious materials. Once a site has been located, a mining ship or station is set up to mine the area
- There are two predominant forms of mineral extraction being considered for full scale operations: continuous-line bucket system (CLB) and the hydraulic suction system.

- The CLB system is the preferred method of nodule collection. It operates much like a conveyor-belt, running from the sea floor to the surface of the ocean where a ship or mining platform extracts the desired minerals, and returns the tailings to the ocean.
- Hydraulic suction mining lowers a pipe to the seafloor which transfers nodules up to the mining ship. Another pipe from the ship to the seafloor returns the tailings to the area of the mining site.